1. Executive Summary

CnEL India proposes to design and develop the **Malus Lens** web platform — a mobile-first, AI-powered apple quality assessment system for cold-storage operators. The solution includes a public marketing landing page, an authenticated product web application, and a backend API that integrates with the client's existing Python ML model running on GPU inference infrastructure.

The platform will enable workers to capture or upload photographs of apple trays, submit them for ML analysis, receive grade-wise counts and percentages, calculate estimated grade-wise weights, and generate a professionally branded PDF report — all without permanent server-side storage of images or reports.

2. Scope Overview — 3 Phases

PHASE 1 — API + ML Integration | Week 1

- REST API for image submission, analysis trigger, and results retrieval
- Integration with client's existing Python ML model (GPU inference)
- Weight calculation (Total Sampled Weight, Estimated A/B Grade Weight)
- Temporary image lifecycle (upload → process → delete)
- Auth system: registration approval, one-device session, 7-day persistent login, forgot password
- Admin APIs: registration approval, org/user management, subscription activation & expiry
- Error handling, retry support, processing status endpoints

Deliverable: Working API that accepts photos + lot inputs and returns grade results + calculated weights.

PHASE 2 — Product Web App (Login + Analysis) | Week 2

- Login / Registration (Organisation, User Name, Email/Mobile, Password; admin approval required)
- Persistent login (7 days) + one-device restriction + logout
- Dashboard with primary "New Analysis" action (no report history)
- Analysis flow: Lot Number, Crates Sampled, Average Weight → 1–10 photos (camera/upload) → review → submit
- Processing state + Results page (counts, percentages, estimated weights)
- Branded PDF report generation (download/share only; no server storage)
- Subscription expiry enforcement (login allowed, analysis blocked)
- Mobile-first UX (large controls, camera access, clear feedback)

Deliverable: Fully functional authenticated analysis app connected to the API/ML.

PHASE 3 — Public Landing Page | Week 3

• Hero section (logo, tagline “Apple Quality. Assessed in Seconds.”, Book a Meeting / Use Malus Lens CTAs)

• Sections: About, Problem, Benefits, Accuracy (placeholder), Product Showcase, Feedback (placeholder), Vision, Contact

• Book a Meeting / Demo integration (Calendly or Google Calendar)

• Navigation + Footer (branding, links, legal placeholders)

• Responsive, mobile-first design; cohesive brand identity (white base, red/dark-red, navy/charcoal)

Deliverable: Live public landing page that drives demos and links to the product app.

3. Assumptions & Notes

- Client provides the ready Python ML model, API contract/details, and brand assets (logo, colour palette).
- No online payment gateway, no report history, no permanent image storage, no usage limits (as per PRD).
- GPU infrastructure selection and any third-party booking tool accounts will be coordinated with the client.
- Admin panel for registration, users, and subscriptions is included within Phase 1 & 2.
- Hosting / client server access for final deployment to be provided by the client.
- Minor content/copy changes during development are included; major scope changes will be quoted separately.